

U20CAM-1080P-WDR UserManual



1. General Description:

Conventional cameras often struggle in high-contrast lighting conditions, resulting in overexposed highlights, dark subjects, or loss of image details. Common scenarios include shooting from indoors toward bright outdoor light, nighttime headlights, and outdoor environments with rapidly changing illumination.

This camera module is equipped with the PS5268 120dB true HDR/WDR Sensor and advanced multi-exposure fusion technology. It effectively suppresses strong background light while enhancing shadow details, allowing the foreground subject to remain clear and visible even in backlight, nighttime, or high dynamic range environments. The result is sharper, more balanced, and detail-rich image quality under challenging lighting conditions.

2. Features

1. **【Plug & Play UVC Camera】**Native UVC compliant USB 2.0 camera module with driver-free setup for Windows, Mac OS, Linux, Ubuntu and Android. Supports up to 1080P@30fps output in YUY2 and MJPEG formats.
2. **【True HDR/WDR Camera】**120dB high dynamic range imaging delivers clear and balanced video in challenging lighting conditions, including backlight, strong contrast and dim environments while preserving highlight and shadow details. A cost-effective alternative to AR0230 based camera modules.
3. **【 Wide-Angle with Microphone】** Fov(D) = 121 degrees. Ultra-wide field of view captures more area with less distortion. Integrated low-noise MEMS microphone provides clear and natural audio for video calls, AI vision and embedded applications.
4. **【OEM-Ready Design】** 32x32mm size with 4x M2 holes. You also could buy the matching metal housings on our Amazon shop separately.
5. **【Compliance And Safety】** RoHS & REACH-SVHC compliant, tested by accredited labs

3. Technical Specification

Sensor	Name	PS5268
	Size	1/2.7 Inch
	Resolution	1920*1080
	Image Color	Colours
	Shutter type	rolling shutter
Lens	F(N) /Aperture	2.2
	Focal Length	5.1
	TV DISTORTION	<-18%
	CRA	12 Degree
	Lens Diameter	14mm
	Lens Seat Spacing	18mm
	FOV-D	121 Degree
	FOV-H	101 Degree
PCBA	Size	32*32 mm
	Layers	4
	Mounting Holes	4 counts, Diameter 2.2mm
	Trigger	None
	Strobe Flash	None
	Night vision	None
	Microphone	Yes
	ESD/EMI	Yes
Interfac es	Communication Interface	USB 2.0 High-Speed
	Cable Length	1 M
Software	Device Name	Innomaker-U20CAM-1080p-WDR
	VID/PID	0x0C45/0x6368
	Communication Standard	UVC 1.0.0
	Supported format	YUY2, MJPEG

	Supported Resolution Supported Frame rate	MJPEG Output resolutions: 1920*1080 @30fps 1280*1024 @30fps 1280x960 @30fps 1280x720 @30fps 1024x768 @30fps 960x720 @30fps 800x600 @30fps 640x480 @30fps 640x360 @30fps 320x240 @30fps YUY2 Output resolutions: 1920*1080 @ 5fps 1280*1024 @ 5fps 1280x960 @ 5fps 1280x720 @ 15fps 1024x768 @ 5fps 960x720 @ 15fps 800x600 @ 20fps 640x480 @ 30fps 640x360 @ 30fps 320x240 @30fps
	Auto Parameters	White Balance Exposure
	Controllable Parameters	Brightness, Contrast, Hue, Saturation, Sharpness, Gamma, White Balance, Backlight Comp, Gain, Exposure, PowerLine Frequency, Low Light Compensation
	Support System	Windows , Linux , Mac Os, Android

3. Software

UVC (USB Video Class) is a protocol standard for USB cameras, allowing for plug-and-play functionality on various platforms. UVC cameras are widely used because they support high-definition video and work across many operating systems without the need for additional drivers.

<https://github.com/INNO-MAKER/UVC-Camera-Software>

You can refer to the official UVC documentation below, but please note that not all UVC cameras are based on version 1.5. You should check the version you are using and refer to the corresponding documentation.

<https://www.usb.org/document-library/video-class-v15-document-set>

4. WDR/HDR Features

WDR (Wide Dynamic Range) and HDR (High Dynamic Range) technologies improve image quality in challenging lighting conditions. By balancing extremely bright and dark areas in the same scene, the camera can preserve more details in highlights and shadows, reducing overexposure and underexposure. This enables clear and natural imaging in backlit, high-contrast and low-light environments, making it ideal for video conferencing, AI vision, industrial and embedded applications.

True WDR HDR Imaging - See Clear in Any Lighting Condition

120dB Wide Dynamic Range, capture more details in bright and dark areas

Indoor Looking Outside Capture both indoor and outdoor details clearly without overexposure.

Without WDR	Digital WDR	PS5268 True WDR HDR

Car Headlights Suppress headlight glare while retaining road and surrounding details.

Without WDR	Digital WDR	PS5268 True WDR HDR

Doorway Backlight Clear identification of people in backlight environments.

Without WDR	Digital WDR	PS5268 True WDR HDR

4. Case

We designed a aluminum alloy case for the camera module. You could find it on our Amazon shop. If you can't catch it, feel free to let me know.

<https://github.com/INNO-MAKER/U20CAM-CASE>



5. User Manual Version Descriptions

Version	Description	Date	E-mail
V1.0		2026.05.10	support@inno-maker.com sales@inno-maker.com

If you have any suggestions, ideas, codes and tools please feel free to email to me. I will update the user manual and record your name and E-mail in list. Look forward to your letter and kindly share.